Incremental Graph Layout by Stochastic Search: a Routed Objective, a Calibrated Metric, and a Matched-Budget Control

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Abstract

Placing the nodes of a graph in the plane with few edge crossings is NP-complete, so every layout tool runs heuristics, and the heuristics are rarely explicit about three things: what objective they optimize, whether that objective measures the drawing the reader actually sees, and whether the search spends its budget any better than uniform random sampling would. We treat all three. The problem is formulated incrementally: a graph grows by k nodes and the existing drawing is frozen, which keeps the reader's mental map, bounds the search space at $2k$ variables, and contains the classic full redraw as the limit of an anchor term that prices the displacement of frozen nodes. The objective reads the drawing's real medium, orthogonally routed connectors rather than the straight center-to-center segments most objectives score, and the metric is calibrated: the router's shortfall against a reference layout of certified on-screen quality is measured and printed beside every number that depends on it. Four families of stochastic search (hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform random sampling as the matched-budget control) are compared by exact sign tests on paired seed panels, deterministically reproducible to the byte across platforms. The motivating application is the entity–relationship diagram, which mainstream tools still place badly enough that maintainers redraw by hand; a proof of concept places the tables added by eight real consecutive schema migrations of a production database, each with a human-accepted answer, under a pre-registered analysis. The searches beat the deployed-style centroid heuristic on every migration (exact Wilcoxon, Holm-corrected $p = 0.023$); hill climbing, not the genetic algorithm the pre-registration predicted, is the budget-efficient method (it wins every non-tied migration against the GA, $p = 0.016$, and separates from the control by 2000 evaluations on most pairs); and on the half of real migrations that add only one or two tables, no search separates from uniform sampling at all, so the first deployable rule is to count the new nodes before reaching for a search. One general finding: under any objective of this family the searches outscore the human reference, which measures the objective's blindness to what maintainers optimize rather than any superiority of the tool, and we argue this gap is why automatic graph layout has stayed bad in practice.

1 Introduction

For most of two decades one of us maintained the entity–relationship diagram of a production database by hand. Every reverse-engineering of the schema replaced the diagram's positions with the tool's own placement, and restoring a 44-table diagram to readability cost about an hour each time. This is not a defect of one tool. MySQL Workbench's placement algorithm is undocumented (its manual names only the menu item), and its failure mode is on the record: bug #42600 reports a 353-table schema autoplaced with some ninety tables stacked into one pile.

*Independent researcher. Code, data and derived tables: `https://github.com/Anode1/cjitter`.

Commercial tools costing more do not necessarily do better; several draw entity–relationship diagrams with straight diagonal edges, a representation no maintainer draws by hand.

The underlying task is hard in the formal sense: placing the nodes of a graph in the plane with the fewest edge crossings is NP-complete [1], so every tool runs heuristics. Our claim is that the practical failure is not the hardness but three things the heuristics leave unstated. First, the objective: what quantity, exactly, is being improved. Second, the medium: whether that quantity is measured on the drawing the reader sees. A tool that optimizes straight center-to-center segments while rendering orthogonal connectors improves a drawing nobody is shown. Third, the control: whether the search spends its evaluation budget any better than uniform random sampling would. In the graph-drawing literature stochastic methods have been compared with one another on quality and time since Davidson and Harel [2], but we have not found a study that runs the random control at matched budget, and without it the claim "our method improves layout" is unfalsifiable at any fixed cost.

This paper treats all three, on the practical version of the problem, which is incremental: a migration adds k tables to a diagram whose other tables must not move, because a reader who knows where a table sits should still find it there. Section 2 formulates the problem and its one-parameter generalization to the classic full redraw. Section 3 makes the objective read the drawing's real medium and calibrates the metric against the one certified fact available. Sections 4 and 5 describe the search harness and a benchmark of real consecutive migrations with human-accepted answers. Section 6 reports a pre-registered experiment whose refutation clause fired against its own author's prediction, which we regard as the methodology working; Section 7 examines what beating the human reference does and does not mean.

2 The problem

Let G be a graph with a drawing D that assigns each node an axis-aligned rectangle in a bounded canvas, and let G' be a supergraph adding k nodes and their edges. The *incremental placement problem* is to position the k new rectangles, with D frozen, minimizing an objective over the resulting drawing, subject to two hard constraints enforced by construction: no two rectangles overlap, and every rectangle lies on the canvas. The search space is $2k$ real variables however large G is.

The frozen context is not a concession to tractability. The dynamic-graph-drawing literature names the reason the *mental map* [3]: a reader who has learned a drawing owns its geography, and a layout that moves what the reader knows destroys more value than it adds. Database migrations sharpen the constraint into an economy: each migration adds a few tables, the diagram's owner accepts a placement, and the accepted drawing becomes the frozen context of the next migration. A version-controlled schema therefore accumulates ground truth: every revision pair is an instance, and the human's accepted answer for it.

One parameter connects this problem to the classic one. Add to the objective an anchor term $\lambda \sum_i \|p_i - p_i^0\|^2$ over the frozen nodes' displacements from their accepted positions and let them move: $\lambda \to \infty$ recovers the incremental problem, $\lambda = 0$ the full redraw, and intermediate values price the reader's geography in the objective's own units. This paper measures the $\lambda \to \infty$ end, where the practical need is; the continuum is the natural sequel, with thirty years of force-directed practice [8] as the control at the unfrozen end.

3 The objective is the medium

Entity–relationship tools draw orthogonal connectors: polylines at 0 and 90 degrees, leaving a table's border, bending once or twice. An objective over straight center-to-center segments scores penetrations and crossings that do not exist on screen and misses ones that do. We therefore route before we measure. Each edge is routed as the cheapest of 24 orthogonal candidates (two L shapes and Z shapes whose middle segment slides across the channel between the endpoints and steps beyond it), anchored on the facing borders at an attachment point unique to that edge, so no two connectors ever share a segment. Routing is sequential and aware: an edge's candidates pay for crossings against every connector already placed, so each pair of edges is priced exactly once and a route can dodge a connector instead of crossing it. The choice is deterministic, ties resolved by declaration order.

The objective is then $S = 100P + 100C + L$ over the routed drawing: penetration P as the total length of connector lying inside tables it does not join (a length, never a count, because a count is flat under small moves and gives a search nothing to follow), crossings C as a count at the same tier, and routed length L two orders of magnitude below, so length breaks ties and no weight is tuned. Node overlap and canvas containment are not terms: they are enforced by a repair callback before any candidate is scored, so no infeasible drawing can be returned as anyone's best.

A routed metric is itself an approximation of the tool's rendering, so we calibrate it against the one certified fact available: the maintainer's accepted layout achieved zero crossings and zero edges under tables on screen. Our router, applied to that layout, measures 27 crossings and 323 units of penetration; the shortfall is the router's, it is printed beside every number that depends on it, and no comparison against the human reference is read past it. The calibration also disciplines development: each router improvement (border anchors, attachment slots, crossing-aware sequencing) was accepted because this number fell, from 36 and 491 at the first orthogonal router to the present values. The same drawing under the straight-segment model charges the certified layout 20 crossings and 1944 penetration, which quantifies how far the diagonal representation misdescribes what the maintainer actually made.

4 Metrics and statistics

The objective over a placement v of the k new nodes is

$$S(v) \;=\; 100\,P(v) \;+\; 100\,C(v) \;+\; L(v),$$

with P the total length of routed connector lying inside a node it does not join, C the count of proper crossings between routed connectors of disjoint node pairs, and L the total routed length; the weights are tiers two orders of magnitude apart, so length only breaks ties and no weight is tuned. The same S is defined for the straight edge model by replacing the router with the direct segment. Beside S:

- the *feasibility pair* (C, P), ordered lexicographically: the part of the score a reader can verify by looking at the drawing;
- the *calibration* $\chi = (C_{\text{ref}}, P_{\text{ref}})$, the metric applied to a reference layout whose on-screen quality is certified as zero and zero. The gap between χ and $(0, 0)$ is the metric's own error, printed beside every number that depends on it;
- the *transfer gap* $\Delta = S_{\text{routed}}(\hat{v}_{\text{straight}}) - S_{\text{routed}}(\hat{v}_{\text{routed}})$: the routed cost of having optimized the straight model, what scoring the wrong medium costs;
- the *separation budget* B^*: the smallest evaluation budget at which a method's exact one-sided sign test against the matched-budget control reaches $p \le 0.05$ on the shared seed panel. A method that never separates gets $B^* = \infty$, a verdict the layout literature does not risk because it does not run the control.

Within an instance, methods share one seed panel and are judged by the exact one-sided sign test on paired per-seed differences. Across instances, the pre-registered comparisons use the exact

Wilcoxon signed-rank test with Holm correction over the declared family, Hodges– Lehmann estimates with distribution-free intervals for effect sizes, and a minimum detectable effect wherever a null is reported, so a null bounds rather than asserts. Realized: $n = 8$ of 16 consecutive pairs (seven added no tables, one was the pilot), five seeds per method per pair, RESULTS.md beside this file carrying the full battery.

5 The searches and the control

The harness, cjitter,[1] runs four methods behind one fitness interface: hill climbing with restarts, simulated annealing, a genetic algorithm with tournament selection and blend crossover, and uniform random sampling as the matched-budget control. Every evaluation passes through one counter, so no method can spend more than another; the budget is exact by construction, and an internal guard turns any violation into an immediate failure rather than a silently unfair comparison. All four run on the same seed panel, and the verdict against the control is an exact one-sided sign test on the paired per-seed differences: "better" requires the wins to be explainable by a fair coin with probability at most five percent, and anything else prints as "not shown," a failure to demonstrate that says nothing about equality.

Two design choices earn their keep in practice. First, determinism: every trajectory uses integer arithmetic and the two libm calls IEEE 754 computes exactly or rounds correctly (fabs, sqrt); the gaussian step is a sum of uniforms rather than Box–Muller, the annealing schedule uses an arithmetic exponential, and a run reproduces byte for byte across compilers and platforms, which is what lets this paper pin every reported number as a regression test. Second, the control catches what tuning hides. The first genetic algorithm shipped in this harness mutated at a fixed scale, re-scattering whatever the population had converged to, and the control's verdict read "no better than uniform sampling at equal cost." An outside review traced the verdict to the fixed scale; with mutation decaying over the run, the same algorithm became the best method on the label-placement example. Both halves of that episode are the point: a method-versus-method table without a random control cannot distinguish a weak method from a mis-tuned one.

6 The benchmark

The instances are real. One production schema's version history holds seventeen revisions of its MySQL Workbench model, sixteen consecutive pairs, each a migration whose diagram a maintainer restored by hand and accepted on both sides. Per pair, the task is the previous revision's diagram frozen as drawn, the added tables to place, and the maintainer's accepted coordinates in the next revision as the human reference; the median displacement of surviving tables between revisions is recorded per pair, since the human answered in a context that sometimes moved. Seven of the sixteen pairs added no tables (documentation edits, pure re-layouts) and are excluded with that reason; one pair served as the pilot and is excluded from the primary analysis; eight remain. One threat is recorded rather than repaired: matching is by table name, so a migration that renames tables can count renames as additions.

The current pair ships with the harness as its example, anonymized for publication: every table renamed to a letter, business column names neutralized down into the view SQL, the machine identifier that v1-style UUIDs embed scrubbed, and the geometry, which is the ground truth, untouched. The mapping back to the schema is deliberately stored nowhere.

[1] C99, no dependencies beyond libm; the repository carries the suites, the data and every number in this paper.

7 Pre-registered experiment

The analysis was committed, timestamped, before any non-pilot pair was scored: primary comparison, declared families, refutation criterion, and the author's own directional prediction. Per pair and method, five seeds at the shipped budget of 8000 evaluations (with 500, 2000 and 32000 for the efficiency curves), the per-pair value the median over seeds. Across pairs, exact Wilcoxon signed-rank tests with Holm correction within each declared family, Hodges–Lehmann shifts with distribution-free intervals, a Friedman omnibus over the four methods gating the pairwise family, exact McNemar on the solve-count binary (whether the method's median-seed layout routes to zero penetration), and a minimum detectable effect beside every null.

The primary comparison succeeded: the genetic algorithm beats the centroid heuristic (each new table at its neighbours' centroid, the deployed-tool stand-in) on eight of eight migrations, exact Wilcoxon $p = 0.0078$, Holm-corrected 0.023, Hodges–Lehmann shift $-72{,}287$ $[-150{,}515, -20{,}728]$; hill climbing against the centroid reads the same. The Friedman omnibus rejects at $p = 0.00045$ with mean ranks climb 1.44, anneal 2.19, GA 2.69, random 3.69.

Two declared clauses then fired against the experimenters, which we report with the same prominence. The refutation clause: the GA did not separate from uniform random sampling ($p = 0.109$; six wins of eight), because half of real migrations add one or two tables, and at $k \leq 2$ every method including the control reaches the same optimum to five digits. Trivial instances cannot discriminate, and the first deployable rule of this paper is to count the new nodes before reaching for a search. And the author's pre-registered prediction, that the GA would be the most efficient search, was refuted by its own experiment: hill climbing beats the GA on every non-tied migration (seven of seven, exact $p = 0.016$, Holm 0.031), separates from the control by 2000 evaluations on most pairs (seven of eight, against the GA's five of eight), and an exploratory bound finds climbing and annealing indistinguishable to within a tenth of the primary effect. The pilot pair, with $k = 10$, had pointed the other way, GA first; the surviving hypothesis, that recombination pays only when there are enough modules to recombine, is the successor pre-registration, and it will be scored the same way this one was.

8 What the human comparison measures

Under the routed objective the searches outscore the maintainer's accepted layouts on most pairs. Read through the calibration line, this is not a claim that the tool out-draws a person. Much of the human reference's score is the router failing to reproduce connectors the maintainer's tool routed cleanly; and the feasibility pairs disagree with the scores, the searches reaching zero penetration while the human's on-screen crossings stay lower. Person and tool jointly achieved zero and zero; each automated party wins a different half of that, and the router owns the remaining gap.

The general reading is the paper's closing argument. A layout objective of this family sees crossings, penetration and length; a maintainer optimizes semantic grouping, aligned rows and room to grow. Any such objective will therefore beat the human on itself and lose on the screen, and a tool that trusts its objective ships the drawing the reader then spends an hour repairing. That gap, between what layout objectives score and what maintainers accept, is our explanation for why automatic diagram layout has stayed bad in practice despite methods that demonstrably beat their controls: the searches are fine, and the objectives are not yet describing the job. Objectives anchored to certified human layouts, with their approximation error measured and printed, are the repair this paper proposes and practices.

9 Related work

Crossing minimization is NP-complete [1], and orthogonal drawing with bend minimization is a literature of its own [4]. Stochastic search for layout begins in earnest with Davidson and Harel's simulated annealing over aesthetic criteria [2], continues through genetic approaches such as TimGA [5], and remains active in metaheuristic comparisons that report final quality and time. The mental map was named and formalized by Misue, Eades, Lai and Sugiyama [3], and incremental drawing under constraints is an established optimization line: the Constrained Incremental Graph Drawing Problem [6] and its successors, most recently hybridized with graph representation learning [7], place vertices on layers under position-shift constraints over synthetic instance sets. Force-directed practice is surveyed in [8], and graphviz's `neato`, which honors pinned positions, stands as the thirty-year baseline our anchor-term sequel must face.

Against that background, the combination here appears unoccupied: geometric frozen-context placement, an objective that routes the medium before measuring it, a metric calibrated against a certified human layout with its error printed, a matched-budget random control, exact paired inference, and a pre-registration whose refutation clauses were allowed to fire. Each ingredient exists somewhere; we have not found the ensemble.

10 Conclusion

An hour of hand repair per reverse-engineering, for years, was a measurement nobody wrote down. Written down properly, it became: a formulation (frozen context, $2k$ variables, an anchor term to the classic problem), an objective that reads routed connectors and confesses its calibration, a harness in which four searches spend identical budgets against a random control and are judged by exact paired tests, a benchmark of eight real migrations with human-accepted answers, and a pre-registered experiment that promoted the searches over the deployed-style heuristic, demoted the author's favored method, and produced the field's plainest deployable rule: count the new nodes first. The successor questions are declared and waiting: whether recombination's advantage returns with the module count, whether the 2001 per-label climber that started this line holds its own inside the harness that finally names it, and at what objective noise any search stops being worth its budget.

Reproducibility

Every number in this paper regenerates from the repository at the frozen commit: the per-seed data, the extraction and analysis code, the pre-registration with its addenda and this paper's source live together, and the harness's own test suite pins the shipped example's every printed digit. The benchmark geometry derives from a private schema and is carried anonymized; the anonymization mapping is stored nowhere.

References

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